



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X
SUBJECT: PHYSICS

FULL MARKS: 40
DATE: 06.05.2024

General Instructions:

- This paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION A
(Attempt all questions.)

Question 1

[1×7=7]

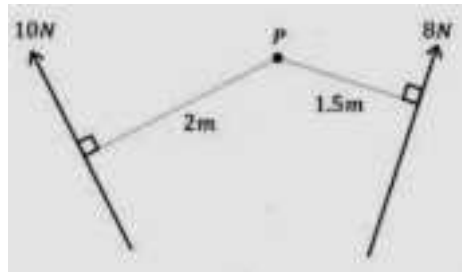
Choose the correct alternative (i), (ii), (iii) or (iv) for each of the questions given below:

- i) Work done against gravity is independent of
 - (a) actual path followed between two points
 - (b) mass of the object
 - (c) height from the ground
 - (d) acceleration due to gravity
- ii) During uniform circular motion,
 - (a) distance -time graph is a curved line
 - (b) displacement-time graph is a straight line
 - (c) distance -time graph is a straight line
 - (d) none of the above
- iii) A force of 20N displaces a body by a distance of 15 cm at an angle 60° to its own direction. The amount of work will be
 - (a) 15J
 - (b) 1.5J
 - (c) 150J
 - (d) 0J
- iv) If the angle between force and displacement is obtuse, then the work done is
 - (a) positive
 - (b) negative
 - (c) zero
 - (d) minimum
- v) Find the odd one out
 - (a) microphone
 - (b) dynamo
 - (c) electric motor
 - (d) nuclear reactor
- vi) The thermocouple changes the heat energy supplied at the _____ junction into electrical energy.
 - (a) cold
 - (b) both
 - (c) hot
 - (d) any one
- vii) If the momentum of a body is increased n times, its kinetic energy increase
 - (a) n times
 - (b) $2n$ times
 - (c) n^3 times
 - (d) n^2 times

Question 2

- i) A stone of mass 'm' is rotated in a circular path with uniform speed by tying a strong string with the help of your hand. Answer the following questions:
 - (a) Is the stone moving with a uniform acceleration?
 - (b) In which direction does the acceleration act?
 - (c) Which force is providing the necessary centripetal force here? [3]
- ii) A flat irregular cardboard lamina is suspended by passing a pin through a narrow hole at its one corner. Draw a diagram to show its position in the state of rest. In the diagram mark the position of suspension by the letter A and centre of gravity by the letter B. [2]
- iii) A body moves along the circumference of a circular track of radius R. What is the displacement of the body when it covers $\frac{3}{4}$ th of its circumference? [2]

- iv) A Racing car is designed to have a broad base and low height-Explain. [2]
 v) The ratio of centre of gravity from the base of two cones, one hollow and one solid is 16:9. Find the ratio of their heights. [2]
 vi) Find the resultant moment of force about P in the following diagram. [2]



SECTION B

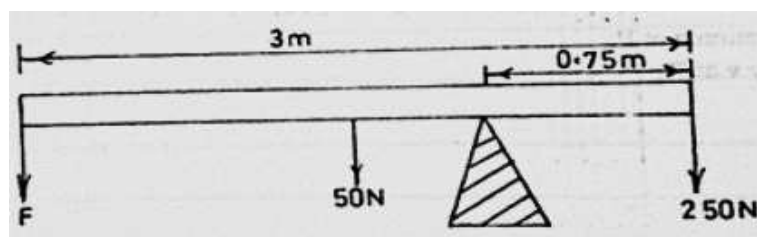
(Attempt all questions.)

Question 3 [3 + 3 + 4 = 10]

- i) A ball of mass 20 g falls from a height and after striking the ground, it rebounds from the ground to a height of 0.8 m. Calculate the velocity of the ball just before it strikes the ground. [take $g = 10 \text{ m/s}^2$]
- ii) Show the variation of two types of mechanical energy for a free falling object graphically. Mention the total energy in the graph.
- iii) (a) Draw a graph to show the work done by force that is directly proportional to displacement. Shade the work done in the diagram.
 (b) State the conditions for equilibrium.

Question 4 [3 + 3 + 4 = 10]

- i) Distinguish between energy and power. What do you understand by dissipation of energy?
- ii) Two forces F and 250 N act on a uniform rigid body of weight 50 N as shown in the diagram below. Calculate the
 (a) clockwise moment about the pivot.
 (b) value of F.



- iii) Draw the graph showing the variation of
 (a) kinetic energy vs. velocity
 (b) momentum vs. kinetic energy